

Capabilities

The following capabilities are the distinct jobs performed by CampusEats, derived from the verbs/actions in the Assignment 1 brief.

No.	Capability	Description
1	Register	Allows users to create an account on the CampusEats platform.
2	Login	Allows users to securely access their CampusEats account.
3	Browse Products	Allows customers to browse available food products from campus vendors.
4	Search Products	Allows customers to search for specific food products.
5	View Product	Allows customers to view product details and prices.
6	Add Product to Cart	Allows customers to add selected food products to their shopping cart.
7	Remove Product from Cart	Allows customers to remove products from their shopping cart.
8	Update Cart Quantity	Allows customers to change the quantity of products in their cart.
9	Place Order	Allows customers to submit their selected products as an order.
10	Make Payment	Allows customers to pay for their food orders online.
11	Cancel Order	Allows customers to cancel an order when cancellation is allowed.
12	Track Order	Allows customers to monitor the status of their food order.
13	Review Product	Allows customers to submit reviews for food products.
14	Add Product	Allows food vendors to add food products to the platform.
15	Update Product	Allows food vendors to update their food product information.
16	Remove Product	Allows food vendors to remove food products from the platform.
17	Manage Orders	Allows food vendors to manage customer orders.
18	Manage Users	Allows administrators to manage platform users.
19	Manage Vendors	Allows administrators to manage food vendors.

CampusEats – Service Contracts

Operations exposed between services, based on the supplied service design diagram. Each contract exposes only the operation, input, output, and caller-visible errors.

1. Order Service — Main Service

Operation	Input	Output	Errors
placeOrder	Student/account reference; selected food items and quantities; delivery address; payment choice.	Accepted order summary, total amount, payment status, delivery status, and current order status.	Invalid item; item unavailable; invalid address; payment declined; order rejected.
getOrder	Order reference and requesting student/account reference.	Order details and current order status.	Order not found; access denied.
cancelOrder	Order reference and requesting student/account reference.	Updated order status confirming cancellation.	Order not found; cancellation not allowed; order already completed.
getOrderStatus	Order reference.	Current order status and relevant payment/delivery state.	Order not found; status unavailable.

2. Catalogue Service — Main Service

Operation	Input	Output	Errors
checkItem	Food-item reference and requested quantity.	Item availability, current price, and restaurant information.	Item not found; item unavailable; catalogue unavailable.
getMenu	Restaurant reference.	Available menu items with names, descriptions, prices, and availability.	Restaurant not found; menu unavailable.
getVendor	Restaurant reference or food-item reference.	Restaurant/vendor information associated with the requested item.	Restaurant not found; item not found; vendor unavailable.
checkAvailability	Food-item reference.	Current availability status.	Item not found; availability unavailable.

3. Payment Service

Operation	Input	Output	Errors
makePayment	Order reference, amount, and selected payment method.	Payment result and payment status.	Payment declined; invalid payment request; payment processing failure.
paymentStatus	Order reference.	Current payment status and transaction result.	Payment not found; status unavailable.

4. Delivery Service

Operation	Input	Output	Errors
assignRider	Order reference, delivery address, and delivery request.	Rider assignment and current delivery status.	Order not eligible; no rider available; assignment failed.
trackDelivery	Order reference.	Current delivery status and latest delivery location.	Delivery not found; tracking unavailable.

5. Notification Service

Operation	Input	Output	Errors
sendNotification	Recipient/account reference, message content, and notification event.	Notification acceptance result and delivery status.	Invalid recipient; unsupported notification; delivery failed.
getNotificationStatus	Notification reference.	Current notification delivery status.	Notification not found; status unavailable.

6. User (Accounts) Service

Operation	Input	Output	Errors
getUser	Student/account reference.	User profile information needed by an authorized caller.	User not found; access denied.
getAddress	Student/account reference and requested delivery-address reference.	Delivery address details.	Address not found; access denied.

Contract boundary: The contracts expose only operations, inputs, outputs, and errors. They do not expose database tables, ID formats, SQL, storage structures, internal algorithms, payment-provider details, rider-selection logic, or notification infrastructure.

CampusEats – Full Specification of *placeOrder*

Central operation of the Order Service. This specification follows the supplied CampusEats service design: Order Service coordinates Catalogue, Payment, Delivery, Notification, and User/Accounts services.

1. Operation Contract

Field	Specification
Operation	placeOrder
Owner	Order Service
Purpose	Accept a student's requested food order, validate the requested items, arrange payment, create the order, request rider assignment, and trigger an order notification.
Inputs	Student/account reference; selected cart/order items and quantities; delivery address; selected payment method; optional order notes.
Output	Accepted order result containing order information, total amount, payment status, delivery status, and current order status.

2. Inputs in Detail

- **Student/account:** identifies the student placing the order.
- **Order items:** the food items requested and their quantities.
- **Delivery address:** where the order should be delivered.
- **Payment choice:** the payment method selected by the student.
- **Optional notes:** any additional order instruction supported by the service.

3. Outputs in Detail

- **Order result:** confirms that the order was accepted.
- **Order information:** the ordered items and calculated total.
- **Payment status:** indicates whether payment was successfully processed.
- **Delivery status:** indicates the current delivery/assignment state.
- **Order status:** indicates the current lifecycle state of the order.

4. Error Cases

Error	When it occurs
Invalid item	A requested food item cannot be accepted by the Catalogue Service.
Item unavailable	A requested item is not currently available.
Invalid address	The supplied delivery address is invalid or cannot be used for delivery.
Payment declined	The Payment Service does not approve the requested payment.
Payment processing failure	The Payment Service cannot complete the payment operation.
No rider available	Delivery cannot assign a rider for the accepted order.
Order rejected	A required business condition prevents the order from being accepted.

5. Internal Processing That Must Be Hidden

- **Catalogue lookup and validation:** Order Service may call *checkItem()* to verify item availability and obtain the information needed to accept the requested items. The Catalogue Service's internal storage and validation logic remain private.
- **Order construction:** Order Service validates the request, calculates the order total, and creates/updates its internal cart and order state. Callers see the result, not the internal representation.
- **Payment coordination:** Order Service may call *makePayment()* and receive *paymentStatus()*. Payment-provider details, transaction processing, and payment internals are hidden.
- **Delivery coordination:** After an accepted order, Order Service may call *assignRider()*. Rider selection, delivery assignment logic, and location handling remain inside Delivery Service.
- **Notification coordination:** Order Service may call *sendNotification()* to notify the student about the order. Notification channels and

delivery mechanisms remain private.

- **Failure handling:** Internal exceptions, retries, private service messages, and partial implementation state are not exposed. The caller receives only the defined output or a defined error.

6. Information Explicitly Hidden from Callers

The caller must not depend on database tables, SQL, storage schemas, internal object structures, ID-generation formats, payment-provider implementation, rider-selection algorithms, notification infrastructure, internal retry logic, or the exact sequence of private service-to-service calls. These are implementation details behind the *placeOrder* contract and may change without changing the public operation.

7. Contract Summary

placeOrder(input) → accepted order result | defined error. The caller knows what information to provide and what result or error to expect, while the Order Service remains free to change how it performs the work internally.

Service Validation

Service	Reachable	Self-contained	Has a Contract
Order Service	Pass — exposes placeOrder, getOrder, cancelOrder, and getOrderStatus.	Pass — manages order creation, order state, and order status.	Pass — operations define inputs, outputs, and errors.
Catalogue Service	Pass — exposes checkItem, getMenu, getVendor, and checkAvailability.	Pass — manages catalogue, menu, item, price, and availability information.	Pass — operations define inputs, outputs, and errors.
Payment Service	Pass — exposes makePayment and paymentStatus.	Pass — manages payment processing and payment status.	Pass — payment operations define inputs, outputs, and errors.
Delivery Service	Pass — exposes assignRider and trackDelivery.	Pass — manages rider assignment and delivery status.	Pass — delivery operations define inputs, outputs, and errors.
Notification Service	Pass — exposes sendNotification and getNotificationStatus.	Pass — manages notifications and delivery status.	Pass — notification operations define inputs, outputs, and errors.
Accounts Service	Pass — exposes getUser and getAddress.	Pass — manages user profile and delivery-address information.	Pass — account operations define inputs, outputs, and errors.

Service	Independent	Loosely Coupled	Fix if Failed
Order Service	Pass — callers use operations without depending on internal implementation.	Pass — communicates with other services through defined operations.	None
Catalogue Service	Pass — internal catalogue storage and validation are hidden.	Pass — services request catalogue information through operations.	None
Payment Service	Pass — provider and transaction-processing details remain internal.	Pass — Order communicates through defined payment operations.	None
Delivery Service	Pass — rider-selection and assignment logic remain internal.	Pass — Order requests delivery operations instead of internal data.	None
Notification Service	Pass — channels and delivery mechanisms remain hidden.	Pass — services use notification operations, not infrastructure.	None
Accounts Service	Pass — internal account storage is hidden from callers.	Pass — services access account information through defined operations.	None